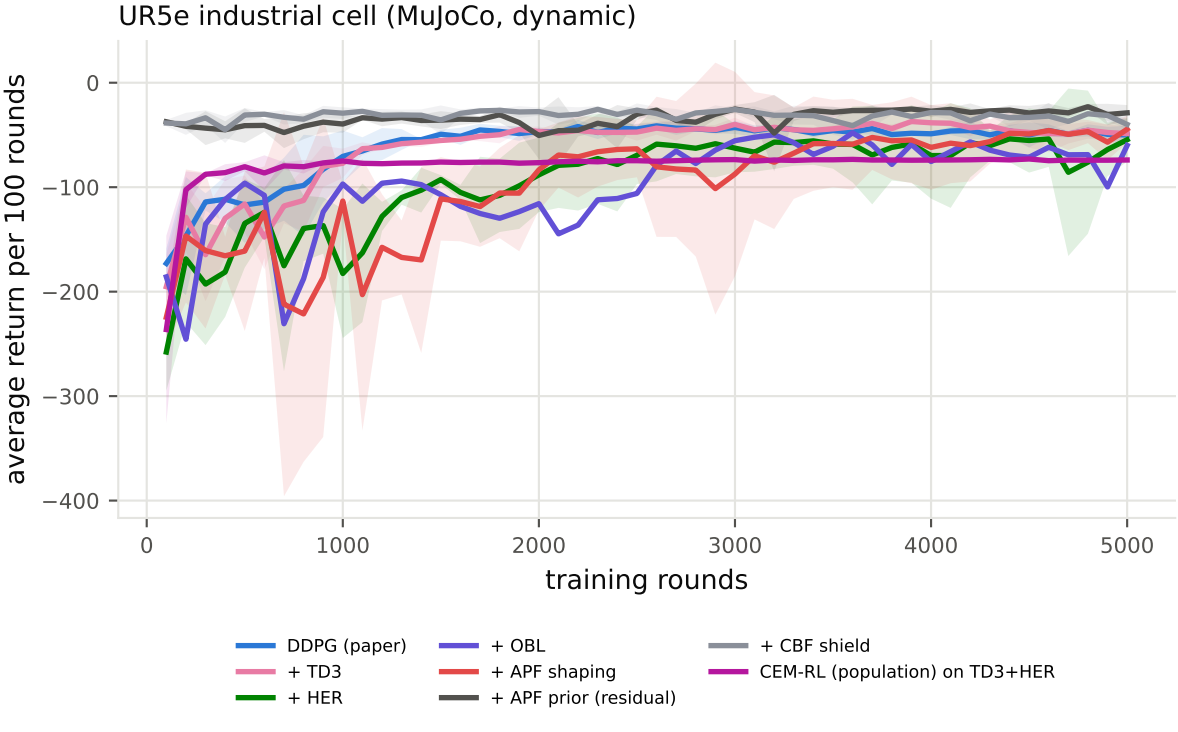
UR5e industrial cell (MuJoCo, dynamic)
average return per 100 rounds
0
−100
−200
−300
−400
0
1000
2000
3000
4000
5000
training rounds
DDPG (paper)
+ TD3
+ HER
+ OBL
+ APF shaping
+ APF prior (residual)
+ CBF shield
CEM-RL (population) on TD3+HER